

SPI MASTER - STATIC TIMING ANALYSIS REPORT

Sky130A | OpenLane v1.0.2 | SS / TT / FF Corners

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SPI MASTER - STATIC TIMING ANALYSIS REPORT

Technology: Sky130A

OpenLane: v1.0.2

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===== SS / MAX CORNER =====

report_tns

tns 0.00

report_wns

wns 0.00

report_worst_slack -max (Setup)

worst slack 2.79

report_worst_slack -min (Hold)

worst slack 0.12

report_clock_skew

Clock clk

Latency CRPR Skew

408/CLK ^

0.67

422/CLK ^

0.59 -0.04 0.04

report_checks -unconstrained

===== Fastest Corner =====

Startpoint: rst (input port clocked by clk)

Endpoint: _409_ (recovery check against rising-edge clock clk)

Path Group: asynchronous

Path Type: max

Corner: Fastest

Fanout Cap Slew Delay Time Description

0.00 0.00 clock clk (rise edge)

0.00 0.00 clock network delay (propagated)

2.00 2.00 v input external delay

0.00 0.00 2.00 v rst (in)

1 0.00 rst (net)

0.00 0.00 2.00 v input2/A (sky130_fd_sc_hd_buf_1)

0.04 0.07 2.07 v input2/X (sky130_fd_sc_hd_buf_1)

3 0.01 net2 (net)

0.04 0.00 2.07 v _355_/A (sky130_fd_sc_hd_buf_4)

0.06 0.13 2.20 v _355_/X (sky130_fd_sc_hd_buf_4)

10 0.05 _199_ (net)

0.06 0.00 2.21 v _356_/A (sky130_fd_sc_hd_buf_4)

0.06 0.14 2.35 v _356_/X (sky130_fd_sc_hd_buf_4)

10 0.06 _200_ (net)

0.07 0.01 2.36 v _358_/A (sky130_fd_sc_hd_inv_2)

0.03 0.05 2.40 ^ _358_/Y (sky130_fd_sc_hd_inv_2)

1 0.00 _003_ (net)

0.03 0.00 2.40 ^ _409_/RESET_B (sky130_fd_sc_hd_dfrtp_4)

2.40 data arrival time

10.00 10.00 clock clk (rise edge)

0.00 10.00 clock source latency

0.06 0.04 10.04 ^ clk (in)

1 0.02 clk (net)

0.06 0.00 10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)

0.05 0.10 10.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)

4 0.04 clknet_0_clk (net)

0.05 0.00 10.14 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd_clkbuf_16)

0.04 0.09 10.24 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd_clkbuf_16)

10 0.04 clknet_2_2_leaf_clk (net)

0.04 0.00 10.24 ^ _409_/CLK (sky130_fd_sc_hd_dfrtp_4)

-0.25 9.99 clock uncertainty

0.00 9.99 clock reconvergence pessimism

0.13 10.12 library recovery time

10.12 data required time

10.12 data required time

-2.40 data arrival time

7.72 slack (MET)

Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: cs (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout Cap Slew Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.06 0.04 0.04 ^ clk (in)
1 0.02 clk (net)
0.06 0.00 0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.11 0.16 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.05 0.00 0.16 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.11 0.26 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3__leaf_clk (net)
0.05 0.00 0.27 ^ _436_/CLK (sky130_fd_sc_hd__dfstp_1)
0.07 0.35 0.61 ^ _436_/Q (sky130_fd_sc_hd__dfstp_1)
2 0.01 net13 (net)
0.07 0.00 0.61 ^ output13/A (sky130_fd_sc_hd__clkbuf_4)
0.09 0.14 0.76 ^ output13/X (sky130_fd_sc_hd__clkbuf_4)
1 0.04 cs (net)
0.09 0.00 0.76 ^ cs (out)
0.76 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock network delay (propagated)
-0.25 9.75 clock uncertainty
0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time

7.75 data required time
-0.76 data arrival time

6.99 slack (MET)

===== Slowest Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.02 0.01 2.01 v rst (in)
1 0.00 rst (net)
0.02 0.00 2.01 v input2/A (sky130_fd_sc_hd__buf_1)
0.11 0.19 2.20 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.11 0.00 2.20 v _355_/A (sky130_fd_sc_hd__buf_4)
0.14 0.37 2.58 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.14 0.00 2.58 v _356_/A (sky130_fd_sc_hd__buf_4)
0.15 0.39 2.97 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.15 0.01 2.98 v _358_/A (sky130_fd_sc_hd__inv_2)
0.06 0.11 3.09 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.06 0.00 3.09 ^ _409_/RESET_B (sky130_fd_sc_hd__dftrtp_4)
3.09 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.13 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.13 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.26 10.35 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 10.35 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.24 10.59 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.04 clknet_2_2__leaf_clk (net)
0.09 0.00 10.59 ^ _409_/CLK (sky130_fd_sc_hd__dftrtp_4)
-0.25 10.34 clock uncertainty
0.00 10.34 clock reconvergence pessimism
0.46 10.80 library recovery time
10.80 data required time

10.80 data required time
-3.09 data arrival time

7.71 slack (MET)

Startpoint: _421_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _427_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description

```

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.13 0.09 0.09 ^ clk (in)
1 0.02 clk (net)
0.13 0.00 0.09 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.29 0.38 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 0.38 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.27 0.66 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3__leaf_clk (net)
0.10 0.00 0.66 ^ _421_/CLK (sky130_fd_sc_hd__dfrrtp_1)
0.16 0.88 1.54 v _421_/Q (sky130_fd_sc_hd__dfrrtp_1)
3 0.01 clk_count[13] (net)
0.16 0.00 1.54 v _205_/A (sky130_fd_sc_hd__or4_1)
0.21 1.34 2.88 v _205_/X (sky130_fd_sc_hd__or4_1)
1 0.00 _097_ (net)
0.21 0.00 2.88 v _209_/A (sky130_fd_sc_hd__nor4_1)
0.74 0.86 3.74 ^ _209_/Y (sky130_fd_sc_hd__nor4_1)
2 0.01 _101_ (net)
0.74 0.00 3.74 ^ max_cap25/A (sky130_fd_sc_hd__buf_1)
0.24 0.49 4.23 ^ max_cap25/X (sky130_fd_sc_hd__buf_1)
3 0.01 net25 (net)
0.24 0.00 4.23 ^ _217_/C (sky130_fd_sc_hd__nand4_1)
0.34 0.43 4.66 v _217_/Y (sky130_fd_sc_hd__nand4_1)
3 0.01 _109_ (net)
0.34 0.00 4.66 v _218_/A (sky130_fd_sc_hd__buf_4)
0.16 0.51 5.17 v _218_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _110_ (net)
0.16 0.01 5.17 v _291_/S (sky130_fd_sc_hd__mux2_1)
0.11 0.75 5.92 v _291_/X (sky130_fd_sc_hd__mux2_1)
1 0.00 _154_ (net)
0.11 0.00 5.92 v hold36/A (sky130_fd_sc_hd__dlygate4sd3_1)
0.11 1.21 7.13 v hold36/X (sky130_fd_sc_hd__dlygate4sd3_1)
1 0.00 net61 (net)
0.11 0.00 7.13 v _292_/A (sky130_fd_sc_hd__clkbuf_1)
0.05 0.19 7.32 v _292_/X (sky130_fd_sc_hd__clkbuf_1)
1 0.00 _067_ (net)
0.05 0.00 7.32 v _427_/D (sky130_fd_sc_hd__dfrrtp_1)
7.32 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.13 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.13 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.26 10.35 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 10.35 ^ clkbuf_2_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.25 10.59 ^ clkbuf_2_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_1__leaf_clk (net)
0.09 0.00 10.60 ^ _427_/CLK (sky130_fd_sc_hd__dfrrtp_1)
-0.25 10.35 clock uncertainty
0.04 10.38 clock reconvergence pessimism
-0.26 10.12 library setup time
10.12 data required time
-----
10.12 data required time
-7.32 data arrival time
-----
2.79 slack (MET)

===== Typical Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.01 0.00 2.00 v rst (in)
1 0.00 rst (net)
0.01 0.00 2.00 v input2/A (sky130_fd_sc_hd__buf_1)
0.05 0.10 2.11 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.05 0.00 2.11 v _355_/A (sky130_fd_sc_hd__buf_4)
0.08 0.20 2.31 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.08 0.00 2.31 v _356_/A (sky130_fd_sc_hd__buf_4)
0.08 0.22 2.53 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.08 0.01 2.53 v _358_/A (sky130_fd_sc_hd__inv_2)
0.04 0.06 2.60 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.04 0.00 2.60 ^ _409_/RESET_B (sky130_fd_sc_hd__dfrrtp_4)
2.60 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)

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0.06 0.15 10.20 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.14 10.34 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.04 clknet_2_2_leaf_clk (net)
0.06 0.00 10.34 ^ _409_/CLK (sky130_fd_sc_hd__dfrtp_4)
-0.25 10.09 clock uncertainty
0.00 10.09 clock reconvergence pessimism
0.22 10.32 library recovery time
10.32 data required time
-----
10.32 data required time
-2.60 data arrival time
-----
7.72 slack (MET)

Startpoint: _421_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _427_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical
Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.08 0.06 0.06 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.16 0.22 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 0.22 ^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.16 0.38 ^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3_leaf_clk (net)
0.06 0.00 0.38 ^ _421_/CLK (sky130_fd_sc_hd__dfrtp_1)
0.08 0.43 0.81 v _421_/Q (sky130_fd_sc_hd__dfrtp_1)
3 0.01 clk_count[13] (net)
0.08 0.00 0.81 v _205_/A (sky130_fd_sc_hd__or4_1)
0.10 0.57 1.38 v _205_/X (sky130_fd_sc_hd__or4_1)
1 0.00 _097_ (net)
0.10 0.00 1.38 v _209_/A (sky130_fd_sc_hd__nor4_1)
0.43 0.47 1.85 ^ _209_/Y (sky130_fd_sc_hd__nor4_1)
2 0.01 _101_ (net)
0.43 0.00 1.85 ^ max_cap25/A (sky130_fd_sc_hd__buf_1)
0.15 0.22 2.07 ^ max_cap25/X (sky130_fd_sc_hd__buf_1)
3 0.01 net25 (net)
0.15 0.00 2.07 ^ _217_/C (sky130_fd_sc_hd__nand4_1)
0.14 0.17 2.25 v _217_/Y (sky130_fd_sc_hd__nand4_1)
3 0.01 _109_ (net)
0.14 0.00 2.25 v _218_/A (sky130_fd_sc_hd__buf_4)
0.09 0.25 2.50 v _218_/X (sky130_fd_sc_hd__buf_4)
10 0.07 _110_ (net)
0.09 0.01 2.50 v _291_/S (sky130_fd_sc_hd__mux2_1)
0.05 0.33 2.84 v _291_/X (sky130_fd_sc_hd__mux2_1)
1 0.00 _154_ (net)
0.05 0.00 2.84 v hold36/A (sky130_fd_sc_hd__dlygate4sd3_1)
0.06 0.58 3.42 v hold36/X (sky130_fd_sc_hd__dlygate4sd3_1)
1 0.00 net61 (net)
0.06 0.00 3.42 v _292_/A (sky130_fd_sc_hd__clkbuf_1)
0.03 0.10 3.52 v _292_/X (sky130_fd_sc_hd__clkbuf_1)
1 0.00 _067_ (net)
0.03 0.00 3.52 v _427_/D (sky130_fd_sc_hd__dfrtp_1)
3.52 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.15 10.20 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_1_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.14 10.34 ^ clkbuf_2_1_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_1_leaf_clk (net)
0.06 0.00 10.35 ^ _427_/CLK (sky130_fd_sc_hd__dfrtp_1)
-0.25 10.10 clock uncertainty
0.02 10.12 clock reconvergence pessimism
-0.10 10.01 library setup time
10.01 data required time
-----
10.01 data required time
-3.52 data arrival time
-----
6.49 slack (MET)

=====
report_checks --slack_max -0.01
=====
===== Fastest Corner =====

No paths found.

===== Slowest Corner =====

No paths found.

===== Typical Corner =====

```

No paths found.

```
=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====
===== Fastest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0__f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1__f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3__f_clk/X 10 12 -2 (VIOLATED)

===== Slowest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0__f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1__f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3__f_clk/X 10 12 -2 (VIOLATED)

===== Typical Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0__f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1__f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3__f_clk/X 10 12 -2 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 3
max cap violation count 0
=====

=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====

===== TT / NOM CORNER =====

=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 2.87

=====
report_worst_slack -min (Hold)
=====
worst slack 0.12

=====
report_clock_skew
=====
Clock clk
Latency CRPR Skew
_408_/CLK ^
0.66
_422_/CLK ^
0.58 -0.04 0.04

=====
report_checks -unconstrained
=====
===== Fastest Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.00 0.00 2.00 v rst (in)
1 0.00 rst (net)
0.00 0.00 2.00 v input2/A (sky130_fd_sc_hd_buf_1)
```

```

0.04 0.07 2.07 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.04 0.00 2.07 v _355_/A (sky130_fd_sc_hd__buf_4)
0.06 0.13 2.20 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.06 0.00 2.20 v _356_/A (sky130_fd_sc_hd__buf_4)
0.06 0.14 2.35 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.06 0.00 2.35 v _358_/A (sky130_fd_sc_hd__inv_2)
0.03 0.04 2.40 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.03 0.00 2.40 ^ _409_/RESET_B (sky130_fd_sc_hd__dftrtp_4)
2.40 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.06 0.04 10.04 ^ clk (in)
1 0.02 clk (net)
0.06 0.00 10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.10 10.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.05 0.00 10.14 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.04 0.09 10.23 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.04 clknet_2_2__leaf_clk (net)
0.04 0.00 10.23 ^ _409_/CLK (sky130_fd_sc_hd__dftrtp_4)
-0.25 9.98 clock uncertainty
0.00 9.98 clock reconvergence pessimism
0.13 10.12 library recovery time
10.12 data required time
-----
10.12 data required time
-2.40 data arrival time
-----
7.72 slack (MET)

Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: cs (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.06 0.04 0.04 ^ clk (in)
1 0.02 clk (net)
0.06 0.00 0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.11 0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.05 0.00 0.15 ^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.11 0.26 ^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3__leaf_clk (net)
0.05 0.00 0.26 ^ _436_/CLK (sky130_fd_sc_hd__dfstp_1)
0.06 0.34 0.61 ^ _436_/Q (sky130_fd_sc_hd__dfstp_1)
2 0.01 net13 (net)
0.06 0.00 0.61 ^ output13/A (sky130_fd_sc_hd__clkbuf_4)
0.09 0.14 0.75 ^ output13/X (sky130_fd_sc_hd__clkbuf_4)
1 0.03 cs (net)
0.09 0.00 0.75 ^ cs (out)
0.75 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock network delay (propagated)
-0.25 9.75 clock uncertainty
0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time
-----
7.75 data required time
-0.75 data arrival time
-----
7.00 slack (MET)

===== Slowest Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.02 0.01 2.01 v rst (in)
1 0.00 rst (net)
0.02 0.00 2.01 v input2/A (sky130_fd_sc_hd__buf_1)
0.11 0.19 2.20 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.11 0.00 2.20 v _355_/A (sky130_fd_sc_hd__buf_4)
0.14 0.37 2.57 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.14 0.00 2.57 v _356_/A (sky130_fd_sc_hd__buf_4)
0.15 0.39 2.97 v _356_/X (sky130_fd_sc_hd__buf_4)

```

```

10 0.06 _200_ (net)
0.15 0.00 2.97 v _358_/A (sky130_fd_sc_hd__inv_2)
0.06 0.11 3.08 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.06 0.00 3.08 ^ _409_/RESET_B (sky130_fd_sc_hd__dfrtp_4)
3.08 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.12 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.26 10.34 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 10.34 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.08 0.24 10.58 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.03 clknet_2_2__leaf_clk (net)
0.09 0.00 10.58 ^ _409_/CLK (sky130_fd_sc_hd__dfrtp_4)
-0.25 10.33 clock uncertainty
0.00 10.33 clock reconvergence pessimism
0.46 10.79 library recovery time
10.79 data required time
-----
10.79 data required time
-3.08 data arrival time
-----
7.71 slack (MET)

Startpoint: _421_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _427_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.12 0.09 0.09 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 0.09 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.29 0.38 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 0.38 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.27 0.65 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3__leaf_clk (net)
0.09 0.00 0.65 ^ _421_/CLK (sky130_fd_sc_hd__dfrtp_1)
0.15 0.87 1.52 v _421_/Q (sky130_fd_sc_hd__dfrtp_1)
3 0.01 clk_count[13] (net)
0.15 0.00 1.52 v _205_/A (sky130_fd_sc_hd__or4_1)
0.20 1.34 2.86 v _205_/X (sky130_fd_sc_hd__or4_1)
1 0.00 _097_ (net)
0.20 0.00 2.86 v _209_/A (sky130_fd_sc_hd__nor4_1)
0.72 0.84 3.70 ^ _209_/Y (sky130_fd_sc_hd__nor4_1)
2 0.01 _101_ (net)
0.72 0.00 3.70 ^ max_cap25/A (sky130_fd_sc_hd__buf_1)
0.24 0.49 4.18 ^ max_cap25/X (sky130_fd_sc_hd__buf_1)
3 0.01 net25 (net)
0.24 0.00 4.18 ^ _217_/C (sky130_fd_sc_hd__nand4_1)
0.33 0.42 4.60 v _217_/Y (sky130_fd_sc_hd__nand4_1)
3 0.01 _109_ (net)
0.33 0.00 4.60 v _218_/A (sky130_fd_sc_hd__buf_4)
0.16 0.50 5.10 v _218_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _110_ (net)
0.16 0.00 5.10 v _291_/S (sky130_fd_sc_hd__mux2_1)
0.10 0.75 5.85 v _291_/X (sky130_fd_sc_hd__mux2_1)
1 0.00 _154_ (net)
0.10 0.00 5.85 v hold36/A (sky130_fd_sc_hd__dlygate4sd3_1)
0.11 1.21 7.05 v hold36/X (sky130_fd_sc_hd__dlygate4sd3_1)
1 0.00 net61 (net)
0.11 0.00 7.05 v _292_/A (sky130_fd_sc_hd__clkbuf_1)
0.05 0.19 7.24 v _292_/X (sky130_fd_sc_hd__clkbuf_1)
1 0.00 _067_ (net)
0.05 0.00 7.24 v _427_/D (sky130_fd_sc_hd__dfrtp_1)
7.24 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.12 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.10 0.26 10.34 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.10 0.00 10.34 ^ clkbuf_2_1__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.24 10.59 ^ clkbuf_2_1__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_1__leaf_clk (net)
0.09 0.00 10.59 ^ _427_/CLK (sky130_fd_sc_hd__dfrtp_1)
-0.25 10.34 clock uncertainty
0.04 10.37 clock reconvergence pessimism
-0.26 10.11 library setup time
10.11 data required time
-----
10.11 data required time
-7.24 data arrival time
-----
2.87 slack (MET)

```

===== Typical Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

Fanout Cap Slew Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.01 0.00 2.00 v rst (in)
1 0.00 rst (net)
0.01 0.00 2.00 v input2/A (sky130_fd_sc_hd__buf_1)
0.05 0.10 2.11 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.05 0.00 2.11 v _355_/A (sky130_fd_sc_hd__buf_4)
0.08 0.20 2.31 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.08 0.00 2.31 v _356_/A (sky130_fd_sc_hd__buf_4)
0.08 0.22 2.53 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.08 0.00 2.53 v _358_/A (sky130_fd_sc_hd__inv_2)
0.03 0.06 2.59 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.03 0.00 2.59 ^ _409_/RESET_B (sky130_fd_sc_hd__dftrtp_4)
2.59 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.15 10.20 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.14 10.34 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.03 clknet_2_2_leaf_clk (net)
0.05 0.00 10.34 ^ _409_/CLK (sky130_fd_sc_hd__dftrtp_4)
-0.25 10.09 clock uncertainty
0.00 10.09 clock reconvergence pessimism
0.22 10.31 library recovery time
10.31 data required time

10.31 data required time
-2.59 data arrival time

7.72 slack (MET)

Startpoint: _421_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _427_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

Fanout Cap Slew Delay Time Description

0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.08 0.06 0.06 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 0.06 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.16 0.22 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 0.22 ^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.15 0.37 ^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3_leaf_clk (net)
0.06 0.00 0.38 ^ _421_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.08 0.42 0.80 v _421_/Q (sky130_fd_sc_hd__dftrtp_1)
3 0.01 clk_count[13] (net)
0.08 0.00 0.80 v _205_/A (sky130_fd_sc_hd__or4_1)
0.09 0.57 1.37 v _205_/X (sky130_fd_sc_hd__or4_1)
1 0.00 _097_ (net)
0.09 0.00 1.37 v _209_/A (sky130_fd_sc_hd__nor4_1)
0.42 0.46 1.83 ^ _209_/Y (sky130_fd_sc_hd__nor4_1)
2 0.01 _101_ (net)
0.42 0.00 1.83 ^ max_cap25/A (sky130_fd_sc_hd__buf_1)
0.15 0.22 2.05 ^ max_cap25/X (sky130_fd_sc_hd__buf_1)
3 0.01 net25 (net)
0.15 0.00 2.05 ^ _217_/C (sky130_fd_sc_hd__nand4_1)
0.13 0.17 2.22 v _217_/Y (sky130_fd_sc_hd__nand4_1)
3 0.01 _109_ (net)
0.13 0.00 2.22 v _218_/A (sky130_fd_sc_hd__buf_4)
0.09 0.25 2.46 v _218_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _110_ (net)
0.09 0.00 2.47 v _291_/S (sky130_fd_sc_hd__mux2_1)
0.05 0.33 2.80 v _291_/X (sky130_fd_sc_hd__mux2_1)
1 0.00 _154_ (net)
0.05 0.00 2.80 v hold36/A (sky130_fd_sc_hd__dlygate4sd3_1)
0.06 0.58 3.38 v hold36/X (sky130_fd_sc_hd__dlygate4sd3_1)
1 0.00 net61 (net)
0.06 0.00 3.38 v _292_/A (sky130_fd_sc_hd__clkbuf_1)
0.03 0.10 3.48 v _292_/X (sky130_fd_sc_hd__clkbuf_1)
1 0.00 _067_ (net)

```

0.03 0.00 3.48 v _427_/D (sky130_fd_sc_hd_dfrtp_1)
3.48 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.06 0.15 10.20 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_1_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.06 0.14 10.34 ^ clkbuf_2_1_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
12 0.04 clknet_2_1_leaf_clk (net)
0.06 0.00 10.34 ^ _427_/CLK (sky130_fd_sc_hd_dfrtp_1)
-0.25 10.09 clock uncertainty
0.02 10.11 clock reconvergence pessimism
-0.10 10.01 library setup time
10.01 data required time
-----
10.01 data required time
-3.48 data arrival time
-----
6.53 slack (MET)

=====
report_checks --slack_max -0.01
=====
===== Fastest Corner =====

No paths found.

===== Slowest Corner =====

No paths found.

===== Typical Corner =====

No paths found.

=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====
===== Fastest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

===== Slowest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

===== Typical Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 3
max cap violation count 0
=====

=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====
===== FF / MIN CORNER =====

report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)

```

```

=====
worst slack 2.93
=====
report_worst_slack -min (Hold)
=====
worst slack 0.12
=====
report_clock_skew
=====
Clock clk
Latency CRPR Skew
_408_/CLK ^
0.65
_422_/CLK ^
0.57 -0.04 0.04
=====
report_checks -unconstrained
=====
===== Fastest Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Fastest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.00 0.00 2.00 v rst (in)
1 0.00 rst (net)
0.00 0.00 2.00 v input2/A (sky130_fd_sc_hd_buf_1)
0.04 0.07 2.07 v input2/X (sky130_fd_sc_hd_buf_1)
3 0.01 net2 (net)
0.04 0.00 2.07 v _355_/A (sky130_fd_sc_hd_buf_4)
0.06 0.13 2.20 v _355_/X (sky130_fd_sc_hd_buf_4)
10 0.05 _199_ (net)
0.06 0.00 2.20 v _356_/A (sky130_fd_sc_hd_buf_4)
0.06 0.14 2.35 v _356_/X (sky130_fd_sc_hd_buf_4)
10 0.06 _200_ (net)
0.06 0.00 2.35 v _358_/A (sky130_fd_sc_hd_inv_2)
0.03 0.04 2.39 ^ _358_/Y (sky130_fd_sc_hd_inv_2)
1 0.00 _003_ (net)
0.03 0.00 2.39 ^ _409_/RESET_B (sky130_fd_sc_hd_dfrtp_4)
2.39 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.06 0.04 10.04 ^ clk (in)
1 0.02 clk (net)
0.06 0.00 10.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.05 0.10 10.14 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4 0.04 clknet_0_clk (net)
0.05 0.00 10.14 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.04 0.09 10.23 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
10 0.03 clknet_2_2_leaf_clk (net)
0.04 0.00 10.23 ^ _409_/CLK (sky130_fd_sc_hd_dfrtp_4)
-0.25 9.98 clock uncertainty
0.00 9.98 clock reconvergence pessimism
0.13 10.11 library recovery time
10.11 data required time
-----
10.11 data required time
-2.39 data arrival time
-----
7.72 slack (MET)

Startpoint: _436_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: cs (output port clocked by clk)
Path Group: clk
Path Type: max
Corner: Fastest

Fanout Cap Slew Delay Time Description
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.06 0.04 0.04 ^ clk (in)
1 0.02 clk (net)
0.06 0.00 0.04 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.05 0.11 0.15 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4 0.04 clknet_0_clk (net)
0.05 0.00 0.15 ^ clkbuf_2_3_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.04 0.11 0.26 ^ clkbuf_2_3_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
12 0.04 clknet_2_3_leaf_clk (net)
0.04 0.00 0.26 ^ _436_/CLK (sky130_fd_sc_hd_dfstp_1)
0.06 0.34 0.60 ^ _436_/Q (sky130_fd_sc_hd_dfstp_1)
2 0.01 net13 (net)
0.06 0.00 0.60 ^ output13/A (sky130_fd_sc_hd_clkbuf_4)
0.09 0.14 0.74 ^ output13/X (sky130_fd_sc_hd_clkbuf_4)
1 0.03 cs (net)
0.09 0.00 0.74 ^ cs (out)
0.74 data arrival time

```

```
10.00 10.00 clock clk (rise edge)
0.00 10.00 clock network delay (propagated)
-0.25 9.75 clock uncertainty
0.00 9.75 clock reconvergence pessimism
-2.00 7.75 output external delay
7.75 data required time
-----
7.75 data required time
-0.74 data arrival time
-----
7.01 slack (MET)
```

===== Slowest Corner =====

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description

```
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.02 0.01 2.01 v rst (in)
1 0.00 rst (net)
0.02 0.00 2.01 v input2/A (sky130_fd_sc_hd__buf_1)
0.11 0.19 2.20 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.11 0.00 2.20 v _355_/A (sky130_fd_sc_hd__buf_4)
0.14 0.37 2.57 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.14 0.00 2.57 v _356_/A (sky130_fd_sc_hd__buf_4)
0.15 0.39 2.96 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.15 0.00 2.96 v _358_/A (sky130_fd_sc_hd__inv_2)
0.05 0.10 3.07 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.05 0.00 3.07 ^ _409_/RESET_B (sky130_fd_sc_hd__dftrtp_4)
3.07 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.12 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.26 10.34 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.09 0.00 10.34 ^ clkbuf_2_2__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.08 0.24 10.57 ^ clkbuf_2_2__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.03 clknet_2_2__leaf_clk (net)
0.08 0.00 10.57 ^ _409_/CLK (sky130_fd_sc_hd__dftrtp_4)
-0.25 10.32 clock uncertainty
0.00 10.32 clock reconvergence pessimism
0.46 10.78 library recovery time
10.78 data required time
-----
10.78 data required time
-3.07 data arrival time
-----
7.71 slack (MET)
```

Startpoint: _421_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _427_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Slowest

Fanout Cap Slew Delay Time Description

```
-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock source latency
0.12 0.08 0.08 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 0.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.29 0.37 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.09 0.00 0.37 ^ clkbuf_2_3__f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.27 0.64 ^ clkbuf_2_3__f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_3__leaf_clk (net)
0.09 0.00 0.64 ^ _421_/CLK (sky130_fd_sc_hd__dftrtp_1)
0.15 0.86 1.50 v _421_/Q (sky130_fd_sc_hd__dftrtp_1)
3 0.01 clk_count[13] (net)
0.15 0.00 1.50 v _205_/A (sky130_fd_sc_hd__or4_1)
0.20 1.33 2.83 v _205_/X (sky130_fd_sc_hd__or4_1)
1 0.00 _097_ (net)
0.20 0.00 2.83 v _209_/A (sky130_fd_sc_hd__nor4_1)
0.72 0.81 3.65 ^ _209_/Y (sky130_fd_sc_hd__nor4_1)
2 0.01 _101_ (net)
0.72 0.00 3.65 ^ max_cap25/A (sky130_fd_sc_hd__buf_1)
0.23 0.48 4.13 ^ max_cap25/X (sky130_fd_sc_hd__buf_1)
3 0.01 net25 (net)
0.23 0.00 4.13 ^ _217_/C (sky130_fd_sc_hd__nand4_1)
0.33 0.41 4.54 v _217_/Y (sky130_fd_sc_hd__nand4_1)
3 0.01 _109_ (net)
```

```

0.33 0.00 4.54 v _218_/A (sky130_fd_sc_hd__buf_4)
0.15 0.49 5.03 v _218_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _110_ (net)
0.15 0.00 5.03 v _291_/S (sky130_fd_sc_hd__mux2_1)
0.11 0.74 5.77 v _291_/X (sky130_fd_sc_hd__mux2_1)
1 0.00 _154_ (net)
0.11 0.00 5.77 v hold36/A (sky130_fd_sc_hd__dlygate4sd3_1)
0.11 1.20 6.98 v hold36/X (sky130_fd_sc_hd__dlygate4sd3_1)
1 0.00 net61 (net)
0.11 0.00 6.98 v _292_/A (sky130_fd_sc_hd__clkbuf_1)
0.05 0.19 7.17 v _292_/X (sky130_fd_sc_hd__clkbuf_1)
1 0.00 _067_ (net)
0.05 0.00 7.17 v _427_/D (sky130_fd_sc_hd__dfrtp_1)
7.17 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.12 0.08 10.08 ^ clk (in)
1 0.02 clk (net)
0.12 0.00 10.08 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.26 10.34 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.09 0.00 10.34 ^ clkbuf_2_1_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.09 0.24 10.58 ^ clkbuf_2_1_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
12 0.04 clknet_2_1_leaf_clk (net)
0.09 0.00 10.58 ^ _427_/CLK (sky130_fd_sc_hd__dfrtp_1)
-0.25 10.33 clock uncertainty
0.04 10.36 clock reconvergence pessimism
-0.26 10.10 library setup time
10.10 data required time
-----
10.10 data required time
-7.17 data arrival time
-----
2.93 slack (MET)

```

===== Typical Corner =====

```

Startpoint: rst (input port clocked by clk)
Endpoint: _409_ (recovery check against rising-edge clock clk)
Path Group: asynchronous
Path Type: max
Corner: Typical

```

Fanout Cap Slew Delay Time Description

```

-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)
2.00 2.00 v input external delay
0.01 0.00 2.00 v rst (in)
1 0.00 rst (net)
0.01 0.00 2.00 v input2/A (sky130_fd_sc_hd__buf_1)
0.05 0.10 2.11 v input2/X (sky130_fd_sc_hd__buf_1)
3 0.01 net2 (net)
0.05 0.00 2.11 v _355_/A (sky130_fd_sc_hd__buf_4)
0.08 0.20 2.31 v _355_/X (sky130_fd_sc_hd__buf_4)
10 0.05 _199_ (net)
0.08 0.00 2.31 v _356_/A (sky130_fd_sc_hd__buf_4)
0.08 0.22 2.52 v _356_/X (sky130_fd_sc_hd__buf_4)
10 0.06 _200_ (net)
0.08 0.00 2.52 v _358_/A (sky130_fd_sc_hd__inv_2)
0.03 0.06 2.58 ^ _358_/Y (sky130_fd_sc_hd__inv_2)
1 0.00 _003_ (net)
0.03 0.00 2.58 ^ _409_/RESET_B (sky130_fd_sc_hd__dfrtp_4)
2.58 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.06 0.15 10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_2_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
0.05 0.13 10.33 ^ clkbuf_2_2_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
10 0.03 clknet_2_2_leaf_clk (net)
0.05 0.00 10.33 ^ _409_/CLK (sky130_fd_sc_hd__dfrtp_4)
-0.25 10.08 clock uncertainty
0.00 10.08 clock reconvergence pessimism
0.22 10.30 library recovery time
10.30 data required time
-----
10.30 data required time
-2.58 data arrival time
-----
7.72 slack (MET)

```

```

Startpoint: start (input port clocked by clk)
Endpoint: _450_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max
Corner: Typical

```

Fanout Cap Slew Delay Time Description

```

-----
0.00 0.00 clock clk (rise edge)
0.00 0.00 clock network delay (propagated)

```

```

2.00 2.00 ^ input external delay
0.02 0.01 2.01 ^ start (in)
1 0.00 start (net)
0.02 0.00 2.01 ^ input3/A (sky130_fd_sc_hd_buf_1)
0.19 0.18 2.19 ^ input3/X (sky130_fd_sc_hd_buf_1)
2 0.02 net3 (net)
0.19 0.00 2.19 ^ _237_/B (sky130_fd_sc_hd_nor2_4)
0.07 0.10 2.29 v _237_/Y (sky130_fd_sc_hd_nor2_4)
9 0.03 _125_ (net)
0.07 0.00 2.29 v _238_/B1 (sky130_fd_sc_hd_a21o_2)
0.10 0.28 2.58 v _238_/X (sky130_fd_sc_hd_a21o_2)
8 0.03 _126_ (net)
0.10 0.00 2.58 v _241_/A2 (sky130_fd_sc_hd_a22o_1)
0.04 0.26 2.84 v _241_/X (sky130_fd_sc_hd_a22o_1)
1 0.00 _090_ (net)
0.04 0.00 2.84 v hold10/A (sky130_fd_sc_hd_dlygate4sd3_1)
0.05 0.57 3.41 v hold10/X (sky130_fd_sc_hd_dlygate4sd3_1)
1 0.00 net35 (net)
0.05 0.00 3.41 v _450_/D (sky130_fd_sc_hd_dfrtp_1)
3.41 data arrival time

10.00 10.00 clock clk (rise edge)
0.00 10.00 clock source latency
0.08 0.05 10.05 ^ clk (in)
1 0.02 clk (net)
0.08 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.06 0.15 10.19 ^ clkbuf_0_clk/X (sky130_fd_sc_hd_clkbuf_16)
4 0.04 clknet_0_clk (net)
0.06 0.00 10.20 ^ clkbuf_2_0_f_clk/A (sky130_fd_sc_hd_clkbuf_16)
0.06 0.14 10.34 ^ clkbuf_2_0_f_clk/X (sky130_fd_sc_hd_clkbuf_16)
14 0.04 clknet_2_0_leaf_clk (net)
0.06 0.00 10.34 ^ _450_/CLK (sky130_fd_sc_hd_dfrtp_1)
-0.25 10.09 clock uncertainty
0.00 10.09 clock reconvergence pessimism
-0.11 9.97 library setup time
9.97 data required time
-----
9.97 data required time
-3.41 data arrival time
-----
6.56 slack (MET)

=====
report_checks --slack_max -0.01
=====
===== Fastest Corner =====

No paths found.

===== Slowest Corner =====

No paths found.

===== Typical Corner =====

No paths found.

=====
report_check_types -max_slew -max_cap -max_fanout -violators
=====
===== Fastest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

===== Slowest Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

===== Typical Corner =====

max fanout
Pin Limit Fanout Slack
-----
clkbuf_2_0_f_clk/X 10 14 -4 (VIOLATED)
clkbuf_2_1_f_clk/X 10 12 -2 (VIOLATED)
clkbuf_2_3_f_clk/X 10 12 -2 (VIOLATED)

=====
report_parasitic_annotation -report_unannotated
=====
Found 0 unannotated drivers.
Found 0 partially unannotated drivers.

=====
max slew violation count 0
max fanout violation count 3
max cap violation count 0

```

```
=====
=====
check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -generated_clocks
=====
```